

BEFORE THE
FEDERAL COMMUNICATIONS COMMISSION
WASHINGTON, D.C. 20554

In the Matter of

Space Exploration Holdings, LLC

ICFS File No.

Application for Authorization of

SAT-LOA-20260108-00016

Non-Geostationary Orbit Satellite

System (Orbital Data Center)

REPLY OF NICKOLAI G. BAKKEN
TO THE CONSOLIDATED OPPOSITION OF
SPACE EXPLORATION HOLDINGS, LLC

This Reply is filed pursuant to 47 C.F.R. Section 25.154(d) and, in the alternative, 47 C.F.R. Sections 1.41 and 1.45, with waiver requested under 47 C.F.R. Section 1.3 to the extent required.

Date: March 22, 2026

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All prior filings in this proceeding, together with all exhibits, appendices, sworn declarations, and preservation clauses, are incorporated by reference in their entirety and remain independently operative. Nothing in this Reply narrows, waives, or supersedes any argument, position, or request for relief previously submitted. Procedural survivability, waiver, and severability provisions are set forth at Appendix I.¹

EXECUTIVE SUMMARY

This Reply examines whether the record before the Commission supports the authorization being requested. It concludes that the record requires extensive supplementation or cure before the Commission can lawfully act, and that on the present record, it cannot.

Section I examines the state of the record as it arrived before this Commission: an Application filed on a placeholder record encoding three satellites for a system of one million, accepted during a partial government shutdown, followed by five independent filings from Petitioner identifying specific defects over twelve days, and a Consolidated Opposition that cured none of them. Section I identifies threshold execution deficiencies in the Opposition under 47 C.F.R.

¹Bakken IO (March 6, 2026): threshold defects including Schedule S encoding, stale Ownership Attachment, absence of bounded technical parameters. Bakken EP (March 9, 2026): environmental petition under 47 C.F.R. Section 1.1307(c) with peer-reviewed science and sworn declaration. Bakken SIO (March 15, 2026): twenty-two structural observations. Bakken CDN (March 15-16, 2026): cross-docket consistency observations. Bakken IR (March 17, 2026): fourteen independent procedural demands under 47 C.F.R. Section 1.41 and 5 U.S.C. Section 555(e), with sworn declaration.

Sections 1.21(d), 1.22, and 1.52, including that the Opposition was signed by an employee of a different legal entity than the Applicant, without verification, without clear demonstrated authorization to represent the Applicant, and without the signature of counsel who was otherwise available and known to parties in the proceeding. Section I requests that the Commission direct the Applicant, which seeks expeditious review, to cure the identified deficiencies within seven days or face striking of the Opposition under Section 1.52.

Even if the Commission exercises its discretion not to strike, the execution posture deprives the Opposition of the evidentiary weight necessary to resolve disputed facts against a certified record. Section II catalogs the un rebutted defects (Table 1), identifies three irreconcilable contradictions on the certified record, and maps the distribution of the record into established, asserted but unverified, and materially unresolved categories.

Section III presents eight determinations the Commission must make before it can act, framed as the questions the record itself raises: *who is the Applicant*, given the undisclosed post-merger ownership structure introducing foreign capital from the Qatar Investment Authority and MGX; *how the environmental certification can be sustained* when the Applicant's own Opposition admits the effects are unknown; *where the technical data* NASA, Petitioner, and other commenters requested can be found; *which certified record the Commission is evaluating*, given that the Schedule S encodes three satellites, the Narrative describes one million, and the Form 312 certifies both as true and correct under 18 U.S.C. Section 1001, and the record attachments were modified after certification; *what the Commission is licensing under Part 25* when the system's primary function is AI computing, and on what congressional authorization; *on what spectrum credibility* the Applicant demands enforcement against competitors while its own Starshield spectrum conduct remains unaddressed; *what the compliance record* actually

shows across four independent federal disclosure regimes; and ***what the Commission required*** for 7,500 Gen2 satellites that the ODC record does not contain for a system more than one hundred times that size.

Section III requests referral to four independent bodies: DDTC for the ITAR compliance question, the Enforcement Bureau for the Starshield unauthorized transmissions and the persisting Section 1.65 violation, the Office of General Counsel for evaluation of Section 1.52 conformity and the adjudicative integrity and character concerns, and the Committee for the Assessment of Foreign Participation for the post-merger ownership structure.

Section IV examines the record the Commission must act on: what it contains, the waivers it cannot support, and the Paperwork Reduction Act limitation that prevents the Commission from waiving the substantive content requirements of an OMB-approved information collection instrument through its general good cause authority. The Schedule S is either complete, in which case the waiver is unnecessary, or incomplete, in which case the certification is false. There is no third position.

The Prayer for Relief offers the Commission a structured range of procedural options calibrated to the condition of the record, from return without prejudice to targeted supplementation to mandatory environmental assessment to hearing designation, together with a seven-day cure demand, enforcement referrals, and denial of the Applicant's requested waivers. Nine Appendices provide full treatment of the authorities and analysis summarized herein.

On the present record, the Application does not satisfy the requirements for an affirmative public interest finding under 47 U.S.C. Section 309(a).

I. THE RECORD BEFORE THE COMMISSION

The Applicant asks the Commission to act expeditiously. Opposition at 24. It requests waiver of the processing round, milestone, surety bond, and Schedule S requirements that constitute the Commission's framework for evaluating applications of this scale. It asks the Commission to trust its institutional reputation and proceed to grant. If that is the posture, then the Commission should evaluate whether the Applicant has provided the Commission with what it needs to act at all.

It has not.

The Application was filed in ICFS on January 30, 2026 on a certified record Form 312 with an encoding of three satellites for a system described as one million. It was accepted within approximately two business days during a partial government shutdown, on a Schedule S displaying an unresolved mandatory validation error, with an Ownership Attachment leaving the majority of equity unattributed and trust beneficiaries unidentified at time of submission, and with an epoch date nearly two years stale. The Applicant described its system as a "first step towards becoming a Kardashev II-level civilization" in its own Narrative.

Petitioner placed five independent filings on this record over twelve days. Each identified specific defects. Each demanded specific responses. Two included sworn declarations and electronic service. The filings were progressively more specific because the record deficiencies they addressed were progressively more confirmed.

The Consolidated Opposition arrived on March 16, 2026, ten days after the first filing by this Petitioner. It contained no curative amendments. No updated Ownership Attachment. No corrected Schedule S. No bounded technical data at fleet scale. No environmental baseline inputs. No response to the ITAR compliance question. No acknowledgment of the Starshield

unauthorized frequency transmissions. No response to the Schedule S encoding discrepancy. What it contained was a request that the Commission proceed "expeditiously" toward grant, a demand that competitors' licenses be revoked, and at footnote 5, the Chairman's own public statement on this proceeding offered as authority for the proposition that a statutory petitioner should not have exercised its rights under 47 U.S.C. Section 309(d).²

An Applicant that demands expeditious action from the Commission while declining to supply the documentation the Commission's own rules require is not cooperating with the regulatory process. It is obstructing it. The Opposition presents threshold execution deficiencies the Commission must resolve before assigning evidentiary weight to its contents.

The Signatory and Authorization Question. The Opposition was signed by Cecilia Tenge-Rietberg, identified as "Senior Satellite Policy Manager, Space Exploration Technologies Corp." The Applicant in this proceeding is Space Exploration Holdings, LLC. The Form 312 was executed by David Goldman, Vice President, Satellite Policy, on behalf of Holdings, LLC. The Opposition was signed by a different individual, with a different title, on behalf of a different legal entity. The record does not contain any document establishing that an employee of Technologies Corp. is authorized to execute pleadings on behalf of Holdings, LLC. Under 47 C.F.R. Section 1.21(d), a corporate employee may act for the corporation only if "duly authorized." Under Section 1.22, the Commission may require any person acting in a representative capacity to show authority. No such showing has been made.³

The Verification Deficiency. Under 47 C.F.R. Section 1.52, when a party is represented by counsel, the pleading shall be signed by at least one attorney of record, and that signature

²See Brendan Carr (@BrendanCarrFCC), X (Mar. 11, 2026 at 10:59 AM ET), cited at Opposition footnote 5. The statement was made on a platform owned by the Applicant's corporate parent during the active pleading cycle of a proceeding in which the Chairman will participate in the final disposition.

³47 C.F.R. Section 1.21(d): "[A] duly authorized corporate officer or employee may act for the corporation in any matter which has not been designated for hearing." 47 C.F.R. Section 1.22: "Any person, in a representative capacity, transacting business with the Commission, may be required to show his authority to act in such capacity."

constitutes a certificate that the contents are well grounded in fact. When a party is not represented by an attorney, the party shall sign and verify the document. The word is "shall." No attorney signed the Opposition. No attorney is identified anywhere in the document. No law firm appears on the Opposition or its Certificate of Service. HWG LLP and William M. Wiltshire have represented the Applicant in satellite matters before this Commission across multiple proceedings. At least two parties in this proceeding, including Petitioner and Amazon/Kuiper, served HWG in connection with filings in this docket. HWG has not disclaimed representation. Yet HWG did not sign the Opposition. And the signatory did not verify it. Whether the Applicant is represented by counsel in this proceeding is a question the record does not answer, and the answer implicates the validity of this filing and any future filings. The Commission should require the Applicant to state its representation posture on the record.⁴

If the Applicant is represented by counsel, the Opposition was required to be signed by counsel and was not. If the Applicant filed without counsel, the signatory was required to verify under oath and did not. Under either reading, the Opposition does not conform with the mandatory execution requirements. There is no third path under the rule. The striking remedy under Section 1.52 is discretionary. Petitioner acknowledges that the Commission may exercise its discretion to accept the Opposition notwithstanding its execution posture. But even if the Commission does not strike the Opposition, the execution deficiencies identified above bear directly on the evidentiary weight the Commission may assign to its contents. An unverified pleading signed by an employee of a different legal entity, without demonstrated authorization,

⁴47 C.F.R. Section 1.52 provides, in relevant part: "The original of all petitions, motions, pleadings, briefs, and other documents filed by any party represented by counsel shall be signed by at least one attorney of record in his individual name." "A party who is not represented by an attorney shall sign and verify the document and state his address." "The signature... by an attorney constitutes a certificate by him that he has read the document; that to the best of his knowledge, information, and belief there is good ground to support it; and that it is not interposed for delay." The rule further provides: "If the original of a document is not signed or is signed with intent to defeat the purpose of this section... it may be stricken as sham and false, and the matter may proceed as though the document had not been filed."

without certification under penalty of perjury, and without the signature of counsel who was available and served by multiple parties in this proceeding, does not carry the reliability necessary to resolve disputed facts against a certified record. The Application was certified under 18 U.S.C. Section 1001. The Bakken EP and Bakken IR were filed under penalty of perjury. The Opposition was not. That asymmetry is not a technicality. It defines what the Commission can rely upon when the representations conflict. The Commission is being asked to resolve factual disputes in the Applicant's favor using advocacy that carries none of the reliability the Commission's own rules and the certified record require.

The Requested Cure. Since the Applicant seeks waiver of the Commission's processing round, milestone, and bond requirements and requests "expeditious" grant, it should have no objection to an expedited compliance timeline. The Applicant is its own corporate entity. It knew the merger was its own transaction. Petitioner provided formal notice on March 6. Petitioner served the Applicant electronically on March 9. The Opposition was filed March 16, acknowledging this Petitioner's filings by name, while remaining silent on every structural defect those filings identified. The Applicant has been on notice for fifteen days as of the date of this Reply. It has otherwise had counsel available throughout. It chose to file without either accountability mechanism the Commission's rules provide. Petitioner requests that the Commission direct the Applicant, within seven days, to: (a) file an updated Ownership Attachment reflecting the current post-merger corporate structure of Space Exploration Holdings, LLC; (b) accompany that filing with a declaration of signatory authority under 47 C.F.R. Section 1.22, identifying the individual authorized to execute filings on behalf of the applicant entity, together with the corporate instrument establishing that authority, and state whether the Applicant is represented by counsel in this proceeding; (c) re-execute or supplement the

Consolidated Opposition with either the signature of counsel of record or verification under oath by the authorized signatory; and (d) submit any technical representations offered to resolve contested issues, including Table 1, in sworn form. Compliance with any of the foregoing constitutes a major amendment under 47 C.F.R. Section 25.116(b), requiring new public notice and a renewed comment period. The Commission should immediately commence a new pleading window upon receipt. If the Applicant does not comply, or if the Commission determines the execution posture was adopted with intent to defeat the purpose of Section 1.52, Petitioner requests the Commission strike the Opposition and proceed on the certified record alone.

The Opposition asks the Commission to "reject calls" that would "monopolize Commission resources." Opposition at 17. The Commission should consider which filing actually burdens its resources: the sworn petitions that attempt to comply with its rules, or the unverified Opposition and stale Application materials that do not.

The question before the Commission is not whether orbital data centers are a good idea. The question is whether the record before this Commission supports the authorization being requested. The Opposition described a system of civilizational importance, then did not mention the word "Kardashev," or those standards, once in thirty-two pages. The Commission should determine which system it is evaluating: the one the Applicant described when it sought authorization, or the one it described when it sought to avoid the standard that description implies.

II. THE RECORD'S CONDITION

A. Unrebutted Defects.

The following structural defects were each raised across prior filings with sufficient specificity to place the Applicant on notice. The Consolidated Opposition failed to provide factual rebuttal or curative amendments for any of them. Each therefore remains uncontested on the record as of the date of this Reply. This list is not all inclusive.

Table 1: Inventory of Unrebutted Threshold Defects

Defect	Requirement	Record / Opposition Posture	Status
Schedule S Encoding	Certified structured data must accurately reflect the proposed system. 18 U.S.C. Section 1001. OMB PWA. Part 25.	Record: Schedule S encodes 3 satellites. Narrative describes up to 1,000,000. Opposition: Acknowledges form cannot describe the system. Provides no amendment. Introduces Table 1 (998,240 satellites) in uncertified form.	Preserved as uncontested.
Section 1.65 Ownership	Self-executing duty to amend within 30 days of material corporate change. 47 C.F.R. Section 1.65.	Record: xAI merger closed Feb. 2, 2026. 30-day deadline expired March 4. Foreign sovereign-affiliated investors (QIA, MGX) entered shareholder base. Opposition: Filed March 16 without updated Ownership Attachment. Did not acknowledge merger.	Ongoing, uncontested rule violation.
Epoch Date	Certified technical data must be temporally consistent with the application. Form 312 Instructions.	Record: Epoch Date on Schedule S is April 1, 2024. Application submitted January 30th, 2026. Form 312 Certified January 31st Opposition: Not addressed.	Preserved as uncontested.
Environmental Categorical Exclusion	Must demonstrate no significant impact. Extraordinary circumstances trigger review. 47 C.F.R. Section 1.1307(c).	Record: Peer-reviewed science (Murphy 2023, Wing 2026) documents atmospheric loading risks. No annualized reentry mass or atmospheric chemistry modeling. Opposition: Admits real-world deployment required to "monitor actual atmospheric effects." Opp. at 16. The Opposition's conformity with 47 C.F.R. Section 1.52 is in question.	Uncontested admission that effects are unknown. Opposition's conformity with 47 C.F.R. Section 1.52 is in question.
Validation Error	ICFS Schedule S must pass mandatory validation before acceptance. 47 C.F.R. Section 25.112.	Record: ICFS Schedule S Review page displays unresolved mandatory validation error. Accepted within 5 days during partial government shutdown. Opposition: Not addressed.	Preserved as uncontested.

Defect	Requirement	Record / Opposition Posture	Status
ITU Coordination	Advance publication and coordination data required. ITU Radio Regulations, Articles 9, 11.	Record: No ITU coordination data. No indication of status of international frequency coordination. No aggregate interference analysis for co-primary Fixed Service operators in the 18.8-19.3 GHz band. ⁵ Opposition: Not addressed.	Preserved as uncontested.
Service Classification	Public notice must accurately describe the system's service type.	Record: Form 312 selected FSS and SRS. Did not select ISS or SOS despite optical ISLs as primary architecture. Opposition: Not addressed.	Preserved as uncontested.
Bounded Technical Parameters	Sufficient data for interference analysis, environmental evaluation, and public review. 47 C.F.R. Section 25.114.	Record: No per-satellite mass, dimensions, disposal-method split, annualized replacement rate, aggregate reentry mass, launch cadence, DAS calculations, or conjunction screening at fleet scale. Opposition: "Is conducting" and "will conduct" detailed analyses. Opp. at 8. NASA independently requested same information. See NASA Comments at 3; Opp. at 13.	Preserved as uncontested. NASA's letter says data is needed.
Signatory Authority	Certification under 18 U.S.C. Section 1001 must be executed by an officer of the applicant entity or other authorized representative. Schedule S / Form 312 Instructions; 47 C.F.R. Section 25.110(a).	Record: David Goldman executed Form 312. No board resolution or delegation letter. Goldman does not appear in Ownership Attachment. The ICFS filing timestamp is January 30, 2026 at 20:51:27; the Form 312 certification is dated January 31, 2026. Opposition: Not addressed.	Preserved as uncontested.

The number and distribution of unresolved discrepancies across independent components of the filing is itself a finding.

B. Three Irreconcilable Contradictions.

Three statements on this record are individually rational but structurally irreconcilable with a certified, complete, evaluable Part 25 application.

⁵ See Comments of Cambium Networks, Inc., ICFS File No. SAT-LOA-20260108-00016 (filed Mar. 6, 2026), at 7-8 (calculating aggregate interference from 500 to 2,000 simultaneously visible co-frequency satellites exceeds PTP 820 receiver sensitivity threshold of -84 dBm in a 60 MHz channel, affecting licensed Part 101 links used for safety-of-life and critical infrastructure communications, and noting that SpaceX's own filing invokes the absence of Article 22 EPFD limits as a feature of its chosen frequency band).

First, the Applicant's phased deployment admission (Opposition at 16) states that models require real-world validation and actual effects must be monitored before scaling. A categorical exclusion presupposes effects are known and insignificant.

Second, the waiver narrative (WAIV at 5) states the form cannot accurately describe the system. The Form 312 certifies the application as true, complete, and correct under 18 U.S.C. Section 1001.

Third, the Opposition minimizes RF usage to near-zero for interference purposes while the Application relies on Part 25 authority predicated on "communication by radio." Each of these tensions requires resolution on the record before the Commission may act.

C. The Distribution of the Record.

Established: The basic procedural history, the identity of the applicant entity, the existence of FAA launch authorizations with a demonstrated flight record, the pleading cycle under DA 26-113, the fact that the Application proposes up to one million satellites between 500 and 2,000 km altitude, that the system's primary described function is AI computing, that NASA filed requesting data from the Applicant the Opposition confirms has not been provided, that at least eight independent parties placed filings, and that the Commission's own Part 100 NPRM acknowledges the framework was not designed for systems of this type. None of the foregoing is disputed. (Appendix B)

Asserted but unverified: Ownership and control (stale, post-merger structure unknown). Technical parameters (no bounded fleet-scale data). Environmental assertions (contradicted by peer-reviewed science). Spectrum compliance representations (unsworn, from an entity with documented unauthorized transmissions on a related constellation).

Materially unresolved: Regulatory classification. Infrastructure dependency across dockets. Character and candor. Cumulative waiver posture. Environmental sufficiency. International treaty compliance. Signatory and certification timeline discrepancies. Issues regarding ICFS's docket on non dot.gov infrastructure, changes to the certified record's attachments without new 312, public notice adequacy, Ex Parte disclosures, use of Applicant's platform by Commissioners during pleading cycle. ⁶

III. THE COMMISSION'S RECORD CHOICES

The Opposition does not limit itself to defending the Application. It advances positions regarding enforcement against other parties, the treatment of existing authorizations, and the availability of waivers. The standards the Applicant urges the Commission to apply necessarily inform the evaluation of the Application itself.

A. Ownership: Who Is the Applicant?

The Applicant bears a self-executing duty under 47 C.F.R. Section 1.65 to amend its application within thirty days of any material change in ownership or control. The xAI merger closed February 2, 2026. The thirty-day deadline expired March 4. No amended Ownership Attachment has been filed. As widely reported and not disputed by any party, foreign sovereign-affiliated investors entered the Applicant's shareholder base through the merger, including the Qatar Investment Authority, the sovereign wealth fund of the State of Qatar, and MGX, a technology investment vehicle based in Abu Dhabi, United Arab Emirates. The

⁶Ownership and control: Appendix A, Section IX. Technical parameters: Table 1 and Appendix F. Environmental assertions: Appendix B. Spectrum compliance: Appendix C. Regulatory classification: Appendix A, Section IV. Infrastructure dependency: Appendix H, Section H.9. Character and candor: Appendix D. Waiver posture: Appendix G. International treaty compliance: Appendix A, Section XI. Signatory discrepancy: Table 1.

introduction of capital affiliated with two foreign sovereign states into the ownership structure of an entity seeking the largest satellite authorization in Commission history, operating infrastructure shared with a classified national security constellation, is precisely the condition the Commission's ownership disclosure framework and the CAFP review process under Executive Order 13913 exist to evaluate. The Applicant's listed trust remains otherwise non-compliant with beneficiary and holding disclosure requirements to give the Commission clear insights into foreign ownership or control of the entities involved with this transaction.

The Applicant is its own corporate entity. It knew the merger was its own transaction. Petitioner provided formal docket notice on March 6 (Bakken IO), March 9 (Bakken EP, electronically served), March 15 (Bakken SIO), March 15-16 (Bakken CDN on ECFS), and March 17 (Bakken IR, electronically served). The Opposition was filed March 16. It acknowledged this Petitioner's March 9th filing. It did not update the Ownership Attachment. It did not acknowledge the merger. No supplemental disclosure has been filed to date.

The National Security dimensions of the AI sector are not hypothetical. In February 2026, the Department of War designated an American AI company a supply-chain risk to national security, a label historically reserved for foreign adversaries, over a policy dispute regarding autonomous weapons and domestic surveillance guardrails. The designation was widely reported and is currently the subject of federal litigation. In the weeks that followed, the Applicant's own AI subsidiary, xAI, was cleared for deployment on the Department of War's classified networks. The Commission need not take a position on that dispute. But its existence confirms that AI companies operating at the intersection of national security infrastructure and supply-chain risk are subject to a level of governmental scrutiny that this record does not reflect.

An Applicant in the position of trust with the Nation's defense and aerospace secrets, whose AI subsidiary now operates on classified military networks, whose satellite constellation provides classified communications services to the United States government, and whose post-merger shareholder base includes capital affiliated with two foreign sovereign states, bears a disclosure obligation with an affirmative showing of the applicant's qualifications under Section 308(b). The record does not reflect that the obligation has been discharged.

The Applicant knows who it is. The Petitioner does not. The Commission, on the current record, does not. The merger closed during the acceptance period and the pleading cycle, and neither the Commission nor any other federal body has evaluated the resulting ownership structure on this record. The identity and qualifications of the applicant are logically prior to every other determination in this proceeding. *See* Appendix A, Section IX.

B. Environment: How Can the Certification remain 'True and Correct'?

The Form 312 certifies, under 18 U.S.C. Section 1001, that the proposed system will have no significant environmental impact. The Applicant's own Opposition states that the actual effects must be determined through deployment and observation, that models require real-world validation, and that adjustments will be made as the system scales. Opposition at 16. A categorical exclusion presupposes that effects are known and insignificant. The Applicant's own words establish that they are neither.

The Bakken EP, filed March 9, 2026, triggered the Bureau's mandatory obligation under 47 C.F.R. Section 1.1307(c). The rule provides that the Bureau "shall" review and consider the environmental concerns raised. Section 1.1313(b) prohibits authorization while such objections remain unresolved. The Bureau has not discharged either obligation.

B1. Section 1.1307, Executive Order 12114, Outer Space Treaty Article IX

Three independent sources require environmental consideration, and the Applicant's defense to NEPA reaches none of them:

1. Section 1.1307(c) is the Commission's own regulation, not a NEPA provision, and remains operative independent of any NEPA amendment.
2. Executive Order 12114 requires federal agencies to evaluate the environmental effects of “major federal actions” on the global commons, including the atmosphere, and operates under the President's constitutional authority; no Commission rulemaking can amend an Executive Order. EO 12114 was not revoked by Executive Order 14154 (January 20, 2025), which contains a specific revocation list that does not include it. The Department of War maintains active implementing regulations at 32 C.F.R. Part 187. NASA complies with EO 12114 for its own space missions, including the Mars Sample Return Campaign, for which NASA prepared a Programmatic Environmental Impact Statement with an EO 12114 compliance checklist. NASA's formal implementing guidance defines global commons to include “the oceans and the atmosphere.” The FCC licenses the same category of space activity that NASA evaluates under EO 12114. The record contains no indication that the Commission has evaluated its EO 12114 obligations with respect to this Application. *See* Appendix B, Section B.7.E.
3. Article IX of the Outer Space Treaty, ratified by the Senate under Article II of the Constitution and binding as the supreme law of the land, requires States Parties to avoid adverse changes to the environment of the Earth resulting from the introduction of extraterrestrial matter. The Applicant's own treaty citation at footnote 30 invokes Article

II but omits both Article VI, which establishes the continuing supervision obligation the Commission's Part 25 authority exists to fulfill, and Article IX.

B1.1. The Scale the Record Does Not Account For.

The Commission cannot satisfy Article IX through a public interest review that contains none of the atmospheric data Article IX's obligation requires it to evaluate. The Commission's own Gen2 environmental analysis relied on ESA studies assuming 450 to 650 tons of reentering satellites per year. One million satellites with a five-year design life implies approximately 200,000 atmospheric reentries per year at steady state. The assumptions are exceeded by orders of magnitude.

B1.2. The Categorical Exclusion Cannot Discharge a Treaty Obligation.

The categorical exclusion under Section 1.1306(a) is a domestic regulatory classification. Article IX of the Outer Space Treaty is a binding international obligation of the United States. The Commission cannot use the former to discharge the latter without ever evaluating whether the latter has been satisfied. The Commission's Part 25 authority exists because Article VI requires the United States to authorize and continuously supervise the space activities of its non-governmental entities. That supervision obligation necessarily includes the Article IX obligation to avoid adverse changes to the Earth's environment from the introduction of extraterrestrial matter. A categorical exclusion that relieves the Commission of evaluating the very environmental consequences its treaty supervision obligation requires it to consider is not a discharge of that obligation. It is an evasion of it.

The question before the Commission is not whether the categorical exclusion applies in the abstract. The question is whether the Commission has evaluated, on this record, whether the Article IX obligation has been satisfied. It has not. The record contains no atmospheric data. The Applicant's own Opposition admits the effects are unknown. NASA's own letter confirms the data is absent. Peer-reviewed science contradicts the environmental certification. The assumptions underlying the Commission's only prior environmental determination on this category of system are exceeded by orders of magnitude. The Commission's Gen2 environmental determination was expressly scale-dependent: it relied on authorizing only 7,500 of 29,988 satellites to avoid the atmospheric loading the parties had estimated. FCC 22-91 at para. 117. An application proposing approximately 200,000 atmospheric reentries per year at steady state is not the same application the Commission evaluated in Gen2. To apply the same conclusion at a scale more than one hundred times larger, without evaluating the Article IX obligation the Commission's own licensing authority exists to fulfill, would require the reasoned departure Fox Television and State Farm demand. See Section III.H.

This analysis does not address the additional supply-chain and strategic materials considerations that a system of this scale would implicate, including the environmental consequences of rare earth extraction, semiconductor fabrication, and launch propellant production at the throughput one million satellites requires, each of which bears independently on the Commission's public interest determination and is preserved for further development as the record warrants.

B2. Categorical Exclusion Waiver Request.

To the extent the Commission determines that the categorical exclusion under 47 C.F.R. Section 1.1306(a) survives the Bakken EP's invocation of Section 1.1307(c), or to the extent any pending rulemaking alters the environmental processing framework before this Application is

resolved, Petitioner requests waiver of Section 1.1306(a) under 47 C.F.R. Section 1.3, with good cause shown by the Executive Order 12114 and Outer Space Treaty obligations and the Bakken Environmental Petition, all bearing on this Application, which the Commission cannot displace by regulation and which independently require environmental consideration prior to authorization. Good cause is further demonstrated by the breadth and specificity of the public interest concerns on this record: the International Astronomical Union, representing more than 12,000 professional astronomers from approximately 90 countries, has stated that even 10% deployment would severely interfere with scientific use of the sky; NASA's own letter calculates an expected casualty count of 100 from the initial fleet, and approximately 20 per year at steady-state replacement rates from reentries at the stated threshold; DarkSky International has filed formal environmental concerns; the American Astronomical Society has filed a Petition to Deny identifying interference with planetary defense, ground-based observatories, and space-based science instruments; and peer-reviewed science documents atmospheric chemistry risks the Applicant's own Opposition admits cannot be quantified without real-world deployment. The Commission's public interest obligation under 47 U.S.C. Section 309(a) does not permit authorization of a system whose environmental consequences are acknowledged as unknown by the Applicant, contradicted by independent science, and opposed by the international scientific community, without first determining whether those consequences warrant assessment.

C. Technical Parameters: Where Is the Data NASA Requested?

The record contains no per-satellite mass, no physical dimensions, no disposal-method split, no annualized replacement rate, no aggregate reentry mass, no launch cadence data, no DAS calculations, and no conjunction screening at fleet scale.

The Opposition cites "NASA comments" throughout as support for the proposition that NASA coordination is underway and proceeding cooperatively. The actual NASA letter filed in this docket does not support that characterization. NASA's letter independently confirms that the record lacks bounded orbital parameters, disposal plan details, deployment timelines, and collision risk data. NASA states that even at 99.9% reliability, up to 1,000 spacecraft could fail in operational orbits. NASA calculates up to 10,000 collisions among retired spacecraft in disposal orbits above 2,000 km over 100 years. NASA observes that reentry of one million spacecraft at the stated casualty threshold could produce an expected casualty count of 100, and recommends full demise be required. NASA states that the milestone waiver creates risks because it leaves NASA unable to plan future missions. NASA requests coordination for any operations crossing the ISS altitude band. In every material respect, NASA's letter supports the deficiencies Petitioner identified and contradicts the Opposition's characterization of them. The Commission should read NASA's actual submission, not the Opposition's summary of it. *See* Appendix H, Section H.3.

The Opposition characterizes the safety and environmental concerns raised by other parties as "baseless fearmongering" from a "familiar cast of competitors." Opposition at 14. NASA is not a competitor. The International Astronomical Union is not a competitor. DarkSky International is not a competitor. NASA's own letter states that the Applicant's stated casualty threshold, applied to one million spacecraft, "can lead to an expectation of casualty of 100 from the reentries of 1 million spacecraft!" and recommends full demise be required. NASA Letter at 3. NASA calculated up to 10,000 collisions among retired spacecraft in disposal orbits, up to 1,000 failures in operational orbits, and stated the Applicant's recycling attestation "cannot be considered viable." NASA Letter at 2-3. The IAU stated even 10% deployment would "severely

interfere with scientific and other uses of orbital space and the sky." The Commission should determine whether the concerns these bodies independently quantified constitute "baseless fearmongering," whether the Opposition's characterization of them on an unverified record is itself a finding the Commission must evaluate.⁷ *See* Appendix H, Section H.3.

D. Schedule S and 312 Certification: Which Record Is the Commission Evaluating?

The certified Schedule S encodes three satellites with an epoch date of April 1, 2024. The Narrative describes one million. The waiver narrative states the form cannot accurately describe the system. The Form 312 certifies the application as true, complete, and correct under 18 U.S.C. Section 1001. Both representations cannot be accurate. In SAT-MOD-20230207-00021, this same Applicant provided supplemental technical information to supplement Schedule S. The Applicant demonstrated it knows how to supplement. It chose not to here. *See* Appendix G.

E. Regulatory Classification: What Exactly Is the Commission Licensing?

The system's primary described function is artificial intelligence computing. Its communications architecture relies "almost entirely" on optical inter-satellite links outside the radiofrequency spectrum. Part 25 governs "communication by radio." 47 U.S.C. Section 153(40). The Opposition minimizes the RF footprint to avoid interference scrutiny while relying on Part 25 authority predicated on communication by radio. Those two positions are in structural

⁷ *See* Letter from Lauren E. Morgan, NASA Representative to the Commercial Transportation Interagency Group, to Marlene H. Dortch, Secretary, FCC, ICFS File No. SAT-LOA-20260108-00016 (NASA Letter); Comments of the International Astronomical Union, ICFS File No. SAT-LOA-20260108-00016 (filed Mar. 3, 2026) (IAU Comments); Comments of DarkSky International, ICFS File No. SAT-LOA-20260108-00016 (filed Mar. 6, 2026) (DarkSky Comments) (formal opposition supported by 290,000 members in over 70 countries documenting light pollution, atmospheric chemistry, and wildlife impacts with peer-reviewed citations, requesting denial unless an Environmental Impact Statement is prepared); Petition to Deny of Amazon Leo, ICFS File No. SAT-LOA-20260108-00016 (filed Mar. 6, 2026); Comments of Cambium Networks, Inc., ICFS File No. SAT-LOA-20260108-00016 (filed Mar. 6, 2026) (original aggregate interference calculations demonstrating receiver threshold exceedance for licensed Part 101 critical infrastructure microwave links). The Opposition addressed none of these filings on the merits.

tension. The Commission must state which governs. To the extent the Commission relies on general Part 25 authority to license a system whose primary function is orbital AI computing at unprecedented scale, the *Major Questions Doctrine* requires clear congressional authorization for that assertion. See Appendix A, Section IV and I.

F. Spectrum: On What Credibility?

The Applicant's opposition demands revocation of competitors' band usage while the Applicant's own Starshield constellation was publicly documented transmitting on unauthorized frequencies in the 2025-2110 MHz band approximately five months before the Opposition was filed. The Bakken IO placed this on the record on 3-6-26. The Opposition did not address it. The standard the Applicant urges the Commission to apply to competitors must apply to it as well. See Appendix C.

G. Character, Candor, Referrals: What Does the Compliance Record Actually Show?

The Commission's 1986 Character Qualifications Policy, 102 F.C.C.2d 1179, identifies compliance history and candor as relevant to applicant qualifications. The Commission has stated that it relies on licensee honesty in a largely self-policing system. *Contemporary Media, Inc. v. FCC*, 214 F.3d 187, 193 (D.C. Cir. 2000). On the present record, the Applicant's character is squarely in question and presently unresolved. Four independent federal compliance regimes impose disclosure obligations or execution requirements bearing on applicant qualifications: ITAR (22 C.F.R. Sections 120.16, 120.54, 127.12), spectrum authorization (47 U.S.C. Sections 312, 503(b)), ownership disclosure (47 C.F.R. Section 1.65), and the mandatory execution requirements of 47 C.F.R. Section 1.52.

G.1. ITAR: Foreign National Access to a Controlled Facility.

Two independent ITAR compliance evaluation moments arise on the federal record, each sufficient standing alone.

The first: the facility visit. Four foreign nationals, two Russian, one South African, and one reported dual citizen Austrian-American, entered SpaceX's Hawthorne headquarters, a facility manufacturing defense articles under USML Categories IV and XV, at the personal invitation of the Applicant's controlling principal.⁸ The presence of any foreign person at such a facility where controlled articles are manufactured or accessible, including lobby access, triggers a mandatory compliance evaluation obligation. 22 C.F.R. Sections 120.16, 120.54, 127.12. The record does not reflect that any such evaluation occurred. The Applicant's controlling principal stated publicly in July 2020 that no such visit had taken place. Federal records released by the Department of Justice in January 2026 demonstrate otherwise, including a contemporaneous email from the visitor stating "thanks for the tour" and a contemporaneous reply from the controlling principal that does not dispute it. The public denial has not been corrected on any record before any agency.

The second: the passport. The American citizen who arranged and participated in the tour was later found, upon execution of a federal search warrant in July 2019, to possess a stamped

⁸ SpaceX's own federal court filing confirms that the entire facility is ITAR-controlled. In its countersuit against the Department of Justice, the company stated: "Every SpaceX employee has access to technology and data controlled by these statutory and regulatory regimes," including ITAR. *Space Exploration Technologies Corp. v. Garland*, No. 7:23-cv-00261, Complaint at 2 (S.D. Tex., filed Sept. 15, 2023). The DOJ's own complaint confirmed that SpaceX restricted facility access to U.S. persons specifically because of ITAR, using rejection codes including "not authorized to work/ITAR ineligible" for applicants who did not meet the company's citizenship requirements. *United States v. Space Exploration Technologies Corp.*, No. 23-INA-00094, Complaint at 3-5 (OCAHO, filed Aug. 24, 2023). SpaceX manufactures articles classified under USML Categories IV (Launch Vehicles) and XV (Spacecraft and Related Articles). 22 C.F.R. Section 121.1. The company's facility-wide access restrictions confirm that SpaceX itself treats the facility boundary, not the production floor, as the ITAR compliance perimeter.

Austrian passport under an alias listing Saudi Arabia as his place of residence and bearing his image. That passport is physical evidence in federal custody. Its existence, contents, and use for international travel are undisputed matters of federal judicial record. When the passport entered the federal record in July 2019, it created an independent ITAR compliance evaluation obligation: the individual who toured the USML facility in 2013 carried a fraudulent identity document with Saudi Arabian dimensions, and Saudi Arabia requires individual export licenses under ITAR. A responsible compliance program would have evaluated whether the original visitor access assessment was adequate and whether a retroactive voluntary disclosure to DDTC was warranted under 22 C.F.R. Section 127.12. The record before this Commission contains no indication that either evaluation was conducted in 2013 when the foreign nationals accessed the facility, or in 2019 when the passport and its Saudi dimensions entered the federal record. *See* Appendix D.

G.2. Spectrum: Unauthorized Transmissions on a Related Constellation.

Under spectrum authorization, the Applicant's Starshield constellation was publicly documented transmitting on unauthorized frequencies in the 2025-2110 MHz band five months before the Opposition was filed. The Bakken IO placed this on the record. The Opposition did not address it.

The record presents three possible explanations:

- (i) authorized classified operations in violation of the allocation framework;
- (ii) unauthorized transmissions from a national security asset;
- (iii) or compromise of a classified constellation's emissions signature.

Each implicates national security. Each requires investigation. All bear directly on the reliability of spectrum compliance representations made by this Applicant for a system one hundred times the size of the constellation in question. *See* Appendix C, Section C.2.

G.3. Ownership: The Persisting Section 1.65 Violation.

Under 47 C.F.R. Section 1.65, the self-executing ownership update obligation was triggered by the February 2, 2026 merger. The thirty-day deadline expired March 4, 2026. No amended disclosure has been filed. The record does not reflect discharge. *See* Appendix A, Section IX.

G.4. Section 1.52: The Opposition's Execution Posture.

The Consolidated Opposition was signed by an employee of a different entity than the Applicant, without verification under oath, and without the signature of counsel who was available and known to every party in the proceeding. Its conformity with 47 C.F.R. Section 1.52 is in question. Petitioner requests a seven-day cure or striking of the Opposition under Section 1.52. *See* Section I. The Opposition's execution posture raises three independent concerns: (1) no record evidence that the signatory can bind the Applicant under Sections 1.21(d) and 1.22; (2) the unverified, unsworn character of the document deprives it of evidentiary weight on disputed facts; and (3) the asymmetry between a certified application under Section 1001 and an uncertified opposition prevents the Commission from resolving conflicts between the two in the Applicant's favor. The full analysis is set forth at Section I of this Reply.

G.5. The Convergence Across Four Regimes.

In each instance, the record reflects conditions that independently trigger disclosure and evaluation obligations. In each instance, the record does not reflect that those obligations were

discharged. The convergence across four regimes addressing the same underlying risk, foreign access to controlled systems and the reliability of representations made to a federal regulator, prevents the affirmative public interest finding Section 309(a) requires.

G.6. Enforcement Referrals (DDTC, EB/NTIA, OGC, CAFP)

The Applicant did not merely open the door to enforcement. It demanded the most severe enforcement remedy the Commission possesses. The Opposition states: "the Commission should not even consider any further arguments from Viasat until the Commission enforces its rules and cancels Viasat's licenses. For over seven years, Viasat brazenly operated GSO systems in NGSO-first spectrum bands without authorization and without regard for the harm it causes to Americans." Opposition at 23. The Applicant asks the Commission to cancel a competitor's licenses for spectrum noncompliance while the Applicant's own Starshield constellation was documented transmitting on unauthorized frequencies five months before the Opposition was filed. The Applicant asks the Commission to punish a competitor for operating "without authorization and without regard" while the Applicant's own Ownership Attachment has not been updated since its corporate identity changed. The Applicant asks the Commission to refuse to "even consider" a competitor's arguments while offering its own arguments in a document that does not fully conform with Section 1.52 and that mischaracterizes NASA's position on the record. The standard the Applicant urges the Commission to apply to its competitors is the standard the Commission must apply to this Application. The record requires nothing less.

The record requires referral to four independent bodies, each addressing a distinct compliance failure documented on this record.

(1) Directorate of Defense Trade Controls (DDTC), U.S. Department of State: For the ITAR compliance question arising from the documented presence of four foreign nationals at the Applicant's USML Category IV and XV facility without evidence of evaluation or disclosure, and the subsequent discovery of a stamped Austrian passport with Saudi Arabian dimensions in federal custody bearing the image of the individual who toured the facility. 22 C.F.R. Section 127.12. *See* Appendix D.

(2) The Commission's Enforcement Bureau, or NTIA as appropriate: For the documented unauthorized transmissions by the Applicant's Starshield constellation in the 2025-2110 MHz band, an uplink-reserved frequency range, in apparent violation of the domestic spectrum allocation framework and ITU Radio Regulations. 47 U.S.C. Sections 312, 503(b). *See* Appendix C. Additionally, for investigation of the Applicant's failure to amend its Ownership Attachment under 47 C.F.R. Section 1.65 following the February 2, 2026 merger, a self-executing obligation that expired March 4, 2026, and which persists unresolved on this record as of the date of this Reply.

(3) The Commission's Office of General Counsel: For evaluation of whether the Opposition conforms with 47 C.F.R. Section 1.52; whether the Ownership Attachment as filed and as it persists satisfies the requirements of 47 C.F.R. Section 25.114(d) and 47 U.S.C. Section 310(b), given that the record does not reflect the post-merger corporate structure, leaves the majority of equity unattributed, and does not identify trust beneficiaries; whether the adjudicative integrity concerns arising from the Chairman's public statements during the pleading cycle warrant referral to the Office of Government Ethics under 5 C.F.R. Part 2635 (*see* Appendix E); and whether the Opposition's characterization of NASA's position is consistent with the actual NASA letter filed in this docket, which independently confirms the deficiencies Petitioner identified and

contradicts the Opposition's representations regarding NASA coordination; and whether the documented pattern of non-disclosure and mischaracterization across four compliance regimes, including the persisting ownership deficiency, warrants evaluation under the Commission's 1986 Character Qualifications Policy, 102 F.C.C.2d 1179.

(4) The Committee for the Assessment of Foreign Participation in the United States Telecommunications Services Sector (CAFP): For evaluation of the post-merger ownership structure, including the introduction of foreign sovereign-affiliated capital from the Qatar Investment Authority, the sovereign wealth fund of the State of Qatar, and MGX, a technology investment vehicle based in Abu Dhabi, United Arab Emirates, neither of which the Applicant has disclosed on this record. Executive Order 13913; 47 C.F.R. Sections 1.5000 *et seq.*

The Applicant opened the door to enforcement by demanding revocation of competitors' authorizations as a matter of policy. The record requires that standard be applied evenhandedly, even strengthened by the mandatory disclosure requirements of actively contested proceedings by applicants before this Commission which afford such public scrutiny.

H. The Commission's Own Framework: What Was Required for 7,500 Satellites?

In the Gen2 First Partial Grant, FCC 22-91 (2022), and the Gen2 Upgrade Order, DA 24-1193 (2024), the Commission established a detailed, condition-laden framework for this same Applicant for this same category of system. The Commission authorized 7,500 of 29,988 requested satellites specifically because of scale concerns. FCC 22-91 at para. 19. It required physical satellite characteristics, DAS logs, NASA coordination, quantitative failure thresholds, EPFD analysis, ITU coordination data, and supplemental Schedule S information as preconditions to grant. The ODC Application satisfies none of these requirements for a system

more than thirty-three times the full Gen2 constellation the Commission chose not to authorize in whole. To grant on the present record, the Commission must depart from its own established framework on every dimension. Any such departure requires a reasoned explanation. *See FCC v. Fox Television Stations, Inc.*, 556 U.S. 502, 515 (2009); *Motor Vehicle Mfrs. Ass'n v. State Farm Mut. Auto. Ins. Co.*, 463 U.S. 29, 43 (1983). *See* Appendix H.

Each determination above is binary. Each requires a stated finding. Silence on any is preserved as an independent ground for reconsideration under 47 U.S.C. Section 405 and judicial review under 5 U.S.C. Section 706.

IV. THE RECORD THE COMMISSION MUST ACT ON

The record is bound by the public docket. This proceeding is designated permit-but-disclose under 47 C.F.R. Section 1.1206. The deficiencies on this record cannot be cured by institutional reputation, by private assurances, or by compliance posture in unrelated dockets. *See Citizens to Preserve Overton Park v. Volpe*, 401 U.S. 402, 420 (1971); *Home Box Office, Inc. v. FCC*, 567 F.2d 9, 54 (D.C. Cir. 1977).

A. What the Record Contains.

The Application encodes three satellites for a system of one million. The Ownership Attachment predates the merger that changed the Applicant's corporate identity. The environmental certification is contradicted by the Applicant's own phased deployment admission. Three irreconcilable contradictions sit on the same certified record. Nine threshold defects remain unrebutted. Four independent compliance regimes reflect conditions triggering disclosure obligations that the record does not show were discharged. The Opposition was signed by an

employee of a different entity than the Applicant, without verification, without demonstrated authorization, and without the signature of counsel who was available and known to every party in the proceeding. The NASA comments the Opposition cites throughout are at odds with the Opposition's characterizations. The single most detailed technical data the Applicant has placed before the Commission exists in a document that satisfies none of the Commission's mandatory execution requirements. And the Commission's own prior orders required bounded physical data, NASA coordination, failure thresholds, EPFD analysis, and ITU coordination as preconditions to authorize a system a fraction of this size. The ODC record contains none of them.

None of this is a matter of characterization. It is a matter of arithmetic and simple validation well within the capabilities of the Applicant. The Applicant served eight parties of record and cited filings from an additional seven organizations in its own footnotes, for a total of at least fifteen independent parties engaged in this proceeding. The Commission's own Part 100 NPRM (FCC 25-69) acknowledges the existing Part 25 framework was not designed for systems of this type. When every independent vector of input points to the same conclusion, the condition of the record is not contested. It is confirmed.

Whatever the Commission does with this Application, it does for every application that follows. The Applicant itself confirmed this principle on March 20, 2026, by filing a letter requesting that the Commission apply the same substantive and procedural standards to a competing orbital data center application.

B. The Waivers the Record Cannot Support.

The Applicant's waiver requests (processing round, milestone, bond, Schedule S) are overbroad, do not cure the structural defects identified herein, and in at least one instance ask the

Commission to waive what cannot be waived. The Schedule S waiver functions as a blanket authority request in tension with FCC 23-73. The milestone and bond waivers remove the only enforcement mechanisms ensuring an authorization of this scale is used, not warehoused, while the Applicant's own Opposition criticizes a competitor for failing its milestones. Petitioner opposes each waiver. See Appendix G.

C. The Paperwork Reduction Act and the Certification Paradox.

The Applicant's Schedule S waiver request fails on an independent ground the Commission cannot cure through its general waiver authority. The Schedule S is an OMB-approved information collection instrument under the Paperwork Reduction Act, 44 U.S.C. Sections 3501-3521, bearing OMB Control Number 3060-0678. The Paperwork Reduction Act framework governs every federal information collection instrument, from the IRS Form 1040 (OMB Control Number 1545-0074), to the Social Security Administration's Application for a Social Security Card, Form SS-5 (OMB Control Number 0960-0066), to the FAA's Airworthiness Certificate application, Form 8130-6 (OMB Control Number 2120-0018). No agency permits an applicant or filer to waive the substantive content requirements of these instruments through a general good cause standard. The Commission's Schedule S occupies the same legal position. The Commission obtained OMB approval by representing that the data fields, including satellite count, orbital parameters, epoch date, and beam information, are necessary for the proper performance of the agency's licensing function. 44 U.S.C. Section 3508. Section 1.3 permits waiver of the Commission's own rules for good cause. It does not permit the Commission to waive the substantive content requirements of an OMB-approved collection without returning to OMB for modified clearance under 44 U.S.C. Section 3507. Good cause cannot override a statutory framework the Commission did not create and does not control.

The Schedule S encodes three satellites for a system of one million. The waiver narrative states the form cannot accurately describe the system. The Form 312 certification, part of the same OMB-approved collection, certifies the application as true, complete, and correct under 18 U.S.C. Section 1001. The certification language was not waived. The Applicant signed it. A certification of completeness on an instrument the Commission has agreed is incomplete is a contradiction no waiver can resolve. The Schedule S is either complete, in which case the waiver is unnecessary, or incomplete, in which case the certification is false. There is no third position.

D. The Applicant Holds the Remedy

The Applicant can file a corrected Schedule S encoding the full constellation or a representative configuration with a specific, non-blanket waiver identifying each rule in conflict. That is supplementation, which the PRA permits. What the PRA does not permit is subtraction: certifying the omission as complete. The waiver should be denied as a matter of law, and the application returned under Section 25.112(a) or held in abeyance until the certified record satisfies the OMB-approved collection requirements.

On the present record, the Application does not satisfy the requirements for an affirmative public interest finding under 47 U.S.C. Section 309(a).

E. Petitioner's Waiver Preservation and Prior Filings.

Petitioner's waiver and preservation requests remain in effect. See Appendix I. All prior prayers for relief remain operative. See Appendix I.

V. PRAYER FOR RELIEF

Petitioner respectfully requests that the Commission align its next procedural step with what the present record is prepared to support.

A. Primary Relief: Return or Dismissal Without Prejudice.

To the extent the Commission determines that the Application was accepted notwithstanding documented deficiencies not cured by waiver, and that any cure would constitute a major amendment under 47 C.F.R. Section 25.116, Petitioner requests that the Commission return the Application under Section 25.112(a), or dismiss it without prejudice to refiling on a complete record.

B. Alternative Relief: Targeted Supplementation and Abeyance.

That the Commission direct the Applicant to cure the documented deficiencies under 47 U.S.C. Section 308(b) and hold processing in abeyance until the record is brought into compliance. Any supplementation altering technical parameters, ownership characteristics, or the potential for environmental impact constitutes a major amendment under Section 25.116(b), requiring new public notice and a renewed comment period.

C. Mandatory Environmental Assessment Prior to Authorization.

The Bakken EP triggered the Bureau's mandatory obligation to review and consider the environmental concerns raised under Section 1.1307(c). Section 1.1313(b) prohibits authorization while the environmental objection remains unresolved. These obligations arise independently from three sources the Commission cannot displace by regulation: (i) Executive

Order 12114, (ii) the Outer Space Treaty (U.S. Const. art. VI, cl. 2), (iii) and Section 1.1307(c) itself. Petitioner requests that the Commission: (i) determine that the Bakken EP triggers the "may have a significant environmental impact" finding; (ii) require the Applicant to prepare an Environmental Assessment under Sections 1.1308 and 1.1311; (iii) withhold authorization until environmental processing is complete; and (iv) reopen the public comment window upon completion. *See* Appendix B, Section B.7. To the extent any Commission rule, categorical exclusion, or pending rulemaking operates to prevent the preparation of an Environmental Assessment for this Application, Petitioner requests waiver of such rule under 47 C.F.R. Section 1.3, with good cause shown by Executive Order 12114, Outer Space Treaty obligations and the Bakken Environmental Petition, all bearing on this Application, which the Commission cannot displace by regulation and which independently require environmental consideration prior to authorization.

D. Hearing Designation for Remaining Material Questions

To the extent substantial and material questions of fact remain that cannot be resolved on the written record, the Commission shall designate the Application for hearing under 47 U.S.C. Section 309(e). The statute is mandatory where the predicate is met. The ownership, classification, character, and environmental questions each independently approach the threshold. Together, they satisfy it. The execution deficiency identified at Section I reinforces this conclusion: where the Applicant's primary responsive pleading does not satisfy mandatory execution requirements, contested factual issues cannot be resolved on the written record in the Applicant's favor, and the statute commands the proceeding the written record cannot support.

E. Section 1.52 Compliance, Cure, and New Pleading Window.

That the Commission direct the Applicant, within seven days, to: (a) file an updated Ownership Attachment; (b) file a declaration of signatory authority under Section 1.22 and state whether it is represented by counsel; (c) re-execute or supplement the Opposition with either the signature of counsel or verification under oath; and (d) submit technical representations, including Table 1 data, in sworn form. Upon receipt, the Commission should commence a new pleading window under Sections 25.116(b) and 25.151. In the alternative, if the Applicant does not comply or if the Commission otherwise determines the execution posture was adopted with intent to defeat the purpose of Section 1.52, that the Commission strike the Opposition and proceed on the certified record alone.

F. Enforcement Referrals. (DDTC, EB/NTIA, OGC, CAFP)

To the extent the record warrants, and without prejudice to any additional referral the Commission determines is appropriate, Petitioner requests the following referrals:

1. **Directorate of Defense Trade Controls (DDTC), U.S. Department of State.** For the ITAR compliance question arising from the documented presence of four foreign nationals at the Applicant's USML Category IV and XV facility without evidence of evaluation or disclosure, and the subsequent discovery of a stamped Austrian passport with Saudi Arabian dimensions in federal custody bearing the image and likeness of the individual who toured the facility. 22 C.F.R. Section 127.12. *See* Appendix D.
2. **The Commission's Enforcement Bureau, or NTIA as appropriate.** For the documented unauthorized transmissions by the Applicant's Starshield constellation in the 2025-2110 MHz band, an uplink-reserved frequency range, in apparent violation of the

domestic spectrum allocation framework and ITU Radio Regulations. 47 U.S.C. Sections 312, 503(b). See Appendix C. Additionally, for investigation of the Applicant's failure to amend its Ownership Attachment under 47 C.F.R. Section 1.65 following the February 2, 2026 merger, a self-executing obligation that expired March 4, 2026, and which persists unresolved on this record as of the date of this Reply. Additionally, for evaluation of whether the Applicant's Schedule S waiver request, which acknowledges the certified instrument cannot accurately describe the proposed system, is reconcilable with the Form 312 certification of that instrument as true, complete, and correct under 18 U.S.C. Section 1001 and consistent with the OMB-approved collection requirements under the Paperwork Reduction Act, 44 U.S.C. Sections 3501-3521, OMB Control Number 3060-0678.

3. **The Commission's Office of General Counsel.** For evaluation of whether the Opposition conforms with 47 C.F.R. Section 1.52; whether the Ownership Attachment as filed and as it persists satisfies the requirements of 47 C.F.R. Section 25.114(d) and 47 U.S.C. Section 310(b), given that the record does not reflect the post-merger corporate structure, leaves the majority of equity unattributed, and does not identify trust beneficiaries; whether the adjudicative integrity concerns arising from the Chairman's public statements during the pleading cycle warrant referral to the Office of Government Ethics under 5 C.F.R. Part 2635 (*see* Appendix E); and whether the Opposition's characterization of NASA's position is consistent with the actual NASA letter filed in this docket, which independently confirms the technical and safety deficiencies Petitioner identified and contradicts the Opposition's representations regarding the status of NASA coordination; and whether the documented pattern of non-disclosure and

mischaracterization across four compliance regimes, including the persisting ownership deficiency, warrants evaluation under the Commission's 1986 Character Qualifications Policy, 102 F.C.C.2d 1179.

4. **The Committee for the Assessment of Foreign Participation in the United States Telecommunications Services Sector (CAFP).** For evaluation of the post-merger ownership structure, including the introduction of foreign sovereign-affiliated capital from the Qatar Investment Authority, the sovereign wealth fund of the State of Qatar, and MGX, a technology investment vehicle based in Abu Dhabi, United Arab Emirates, neither of which the Applicant has disclosed on this record. Executive Order 13913; 47 C.F.R. Sections 1.5000 *et seq.*

G. Denial of Applicant's Waivers.

That the Commission deny the Applicant's requests for waiver of the processing round (Section 25.157), milestone (Section 25.164), surety bond (Section 25.165), and Schedule S (Section 25.114(a)(1)) requirements for the reasons set forth at Section IV of this Reply and Appendix G. That the Commission deny the Applicant's Schedule S waiver request on the independent ground that the Commission's Section 1.3 waiver authority does not extend to waiving the substantive content requirements of an OMB-approved information collection instrument under the Paperwork Reduction Act, 44 U.S.C. Sections 3501-3521, OMB Control Number 3060-0678, without obtaining modified clearance from OMB under 44 U.S.C. Section 3507; and that the Commission direct the Applicant to file a corrected, complete Schedule S and contemporaneous Form 312 certification within no more than thirty days, or in the alternative, return the application under 47 C.F.R. Section 25.112(a) until the certified record satisfies the OMB-approved collection requirements.

H. Post-Pleading Disclosure and Record Integrity.

That the Commission take notice that on March 21, 2026, one day before the close of the reply window, the Applicant publicly disclosed technical parameters for the Orbital Data Center satellites, including per-satellite mass (approximately one metric ton), physical dimensions (exceeding 170 meters when deployed), power architecture (100 kilowatts per satellite, with future satellites scaling to the megawatt range), thermal management specifications (approximately 100 square meter radiator), a custom radiation-hardened chip program (designated "D3"), and a \$20-25 billion joint chip fabrication venture among SpaceX, Tesla, and xAI (designated "Terafab"), with 80% of output directed to space-based applications, none of which appears anywhere on the Commission's record. See Appendix J.

That the Commission determine whether the Applicant's public disclosure of bounded technical and programmatic parameters that it has not filed with the Commission is consistent with its obligations under 47 C.F.R. Section 1.65, its Form 312 certification under 18 U.S.C. Section 1001, and the Commission's 1986 Character Qualifications Policy, 102 F.C.C.2d 1179. That the Commission determine whether the certified record, which encodes three satellites with no physical parameters, remains reconcilable with the Applicant's own public representations regarding the system it is asking the Commission to authorize. That the Commission direct the Applicant to file, within the seven-day cure period identified at Section V.E of this Reply, all technical parameters disclosed at the March 21, 2026 event, in certified form, accompanied by new public notice and a renewed comment period under 47 C.F.R. Sections 25.116(b) and 25.151, so that all parties may evaluate the system the Applicant has described to the world on the record where the Commission needs it.

I. Document Preservation and Pending FOIA Requests.

That the Commission preserve all records in the eleven categories identified in the Bakken IR, Request 11 (Categories A through K), process the three expedited FOIA requests identified therein under 5 U.S.C. Section 552(a)(6)(E), and not permit the routine destruction of any records bearing on the acceptance determination, internal processing standards, or pre-filing communications regarding this Application pending resolution of this proceeding and the FOIA requests, including records maintained by or on behalf of the Commission through any third-party vendor, contractor, or platform operator.

J. General Relief.

Petitioner requests such other and further relief and necessary waiver or reply vehicle as the Commission deems just, proper, and consistent with the public interest, convenience, and necessity.

K. Preservation of Prayers.

To the fullest extent allowable by law, Petitioner preserves the right to seek relief through any available mechanism, including but not limited to: forfeiture proceedings under 47 U.S.C. Section 503(b); revocation proceedings under 47 U.S.C. Section 312; petitions for order to show cause under 47 U.S.C. Section 312(c); requests for investigation under 47 U.S.C. Section 403; formal complaints under 47 U.S.C. Section 208; and any other remedy available through the Commission, the Department of Justice, the Department of State, or any other forum of competent jurisdiction, to the extent the record in this proceeding or any subsequent proceeding warrants such relief in the public interest. Petitioner further preserves arguments arising under

the Anti-Deficiency Act, 31 U.S.C. Section 1341, to the extent the Commission's authorization of this Application commits the United States to aggregate absolute liability under Article II of the Convention on International Liability for Damage Caused by Space Objects that has not been quantified or appropriated for, or any other appropriation activity; and under 50 U.S.C. Section 3091, to the extent classified spectrum operations by the Applicant's related constellation bear on the Commission's public interest determination and the congressional oversight framework applicable to intelligence activities. All prayers for relief in the Bakken IO, Bakken EP, Bakken SIO, Bakken CDN, and Bakken IR remain in full force and effect, incorporated by reference, and independently operative. The fourteen requests in the Bakken IR, the preserved authorities at Appendix A, and all prior prayers are not narrowed, superseded, or waived by this Reply. Silent resolution of any is preserved for reconsideration under 47 U.S.C. Section 405 and judicial review under 47 U.S.C. Section 402(b) and 5 U.S.C. Section 706.

VI. CLOSING

This record does not yet meet the ambition it raises. The Application was incomplete at filing. That condition was documented across five independent instruments. The Opposition confirmed it. The Applicant can prove otherwise. It can file a sworn, updated Ownership Attachment. It can submit bounded technical data at fleet scale. It can provide the environmental baseline inputs peer-reviewed science requires. So far, it has not.

A. Applicants March 21st ODC Event

On March 21, 2026, the evening before this Reply was filed and one day before the close of the reply window, the Applicant's controlling principal held a public event in Austin, Texas, at which SpaceX, Tesla, and xAI jointly disclosed technical, programmatic, and corporate details about the Orbital Data Center system that do not appear anywhere on the Commission's record. The Applicant revealed the physical satellite design, designated the "AI Sat Mini": approximately one metric ton, 100 kilowatts of onboard power, and when unfolded with solar arrays and radiators, larger than the Starship V3 rocket itself. The Applicant announced a custom radiation-hardened "D3" chip, a \$20-25 billion joint chip fabrication facility called "Terafab," and stated that 80% of the facility's output would be directed to space-based applications. The Applicant described future satellites scaling to the megawatt range. None of this information was filed with the Commission. None of it was available to any party during the pleading window.

The Commission's record encodes three satellites with no physical parameters. The Applicant just told the world each satellite is larger than its own rocket. The full inventory of

publicly disclosed but unfiled technical and programmatic details, together with the record and procedural implications, is set forth at Appendix J.

A decision grounded in a fully developed record will be more precise and more stable. A decision that proceeds on the current record inherits every unresolved deficiency that record contains. The conditions for lawful authorization have not been satisfied on the present record.

Unless and until they are, the Commission should act accordingly on behalf of its public interest and national security mandates.

Respectfully submitted,

/s/ Nikolai G. Bakken

Nikolai G. Bakken
Pro Se
Risaralda, Colombia
March 22, 2026

APPENDICES

The following Appendices are filed concurrently with this Reply and are incorporated herein by reference.

Appendix A: Preserved Authorities and Procedural Predicates (Sections I-XIV)

Appendix B: The FAA Reliance Gap, the Commission's Independent Environmental Obligation, and the Outer Space Treaty

Appendix C: The Starshield Spectrum Record

Appendix D: The ITAR Compliance Question and the Epstein Files (Sections D.1-D.10)

Appendix E: ODC Proliferation, Delegated Authority, Sunshine Act, and the Forum This Proceeding Requires

Appendix F: Record Sufficiency Matrix (Twenty-Two Observations)

Appendix G: The Schedule S Waiver as Blanket Authority Request

Appendix H: The Commission's Own Framework Applied to the ODC Application (Sections H.1-H.10)

Appendix I: Waiver and Procedural Survivability

Appendix J: The Terafab disclosure and the record the applicant chose not to file

VERIFICATION

I, Nickolai G. Bakken, declare under penalty of perjury under the laws of the United States of America that I am the Petitioner in this proceeding; that I am a citizen of the United States traveling in Risaralda, Colombia; and that I have read the foregoing Reply and the Appendices filed concurrently herewith.

The factual statements concerning my personal identity, my submissions to ICFS/ECFS including under SAT-LOA-20260108-00016 and their dates in this docket are true and correct to the best of my personal knowledge.

The citations to federal records, administrative dockets, published orders, Department of Justice releases, international treaties, and scientific literature are accurately reproduced and characterized to the best of my knowledge, information, and belief.

The legal arguments, procedural analyses, and inferences drawn from the record are offered as advocacy and argument, not as matters of personal knowledge.

Executed on this 22nd day of March, 2026 in Risaralda, Colombia

/s/ Nickolai G. Bakken

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CERTIFICATE OF SERVICE

I, Nikolai G. Bakken, hereby certify that on March 22, 2026, courtesy copies of the foregoing Reply and Appendices were served by electronic mail on all parties listed below for whom valid email addresses were available from filings in this proceeding.

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Nikolai G. Bakken

March 22, 2026

NOTICE OF DOCUMENT INTEGRITY CERTIFICATION

A Certificate of Document Integrity, including SHA-256 hash values for this Reply and each Appendix filed concurrently herewith, will be filed separately in this docket. That Certificate will confirm the contents of each document as transmitted to the Commission and served on the parties listed above. Any discrepancy between the hash values recorded at time of filing and the documents appearing on the docket is preserved as a record integrity issue for all purposes.

- END OF FILING -